

Advance 2 Project: Financial Statement Dashboard in Power BI

Objective

To visualize and monitor a company's financial health through:

- ❖ Profitability
- ❖ Revenue vs Expenses
- ❖ Assets vs Liabilities
- ❖ Cash Flow over time
- ❖ Financial Ratios and KPIs

Step 1: Prepare the Excel Dataset

Assume the Excel file (Financial_Statements.xlsx) includes **3 sheets**:

A. Income_Statement

| Date | Revenue | COGS | Gross_Profit | Operating_Expense | Net_Income |

B. Balance_Sheet

| Date | Assets | Liabilities | Equity |

C. Cash_Flow

| Date | Cash_Inflows | Cash_Outflows | Net_Cash_Flow |

Optional Columns (Advanced): Department, Account_Type, Category

Step 2: Load Dataset into Power BI

1. Open **Power BI Desktop**
2. Click **Home** → **Get Data** → **Excel**
3. Browse and select your Excel file
4. Load all 3 sheets:
 - ❖ Income_Statement
 - ❖ Balance_Sheet
 - ❖ Cash_Flow

Step 3: Transform and Clean Data (Power Query Editor)

Click “**Transform Data**” and do the following:

- ✓ Data Types:
 - ❖ Set Date columns to **Date**

- ❖ Set numeric columns to **Decimal Number**

✓ Clean/Format:

- ❖ Remove extra spaces or commas
- ❖ Handle missing values
- ❖ Convert text to proper format

✓ Add Custom Columns (if not in Excel):

- ❖ **Gross Margin** = [Gross_Profit] / [Revenue]
- ❖ **Net Profit Margin** = [Net_Income] / [Revenue]
- ❖ **Debt to Equity Ratio** = [Liabilities] / [Equity]
- ❖ **Working Capital** = [Assets] - [Liabilities]

Click **Close & Apply**

Step 4: Create Relationships

1. Go to **Model View**
2. Create a **Date Table**:

DateTable = CALENDAR(MIN(Income_Statement[Date]),
MAX(Income_Statement[Date]))

3. Create relationships:

- ❖ DateTable[Date] → Income_Statement[Date]
- ❖ DateTable[Date] → Balance_Sheet[Date]
- ❖ DateTable[Date] → Cash_Flow[Date]

Step 5: Create Key DAX Measures

Total Revenue = SUM(Income_Statement[Revenue])

Total Net Income = SUM(Income_Statement[Net_Income])

Gross Margin % = DIVIDE(SUM(Income_Statement[Gross_Profit]),
SUM(Income_Statement[Revenue]))

Net Profit Margin % = DIVIDE(SUM(Income_Statement[Net_Income]),
SUM(Income_Statement[Revenue]))

Debt to Equity Ratio = DIVIDE(SUM(Balance_Sheet[Liabilities]),
SUM(Balance_Sheet[Equity]))

Working Capital = SUM(Balance_Sheet[Assets]) - SUM(Balance_Sheet[Liabilities])

Net Cash Flow = SUM(Cash_Flow[Net_Cash_Flow])

Step 6: Create Visualizations (Charts and Graphs)

A. KPI Cards (Top Row)

1. Go to **Visualizations Pane** → **Card**

2. Drag your DAX Measures:

- ❖ Total Revenue
- ❖ Net Income
- ❖ Assets
- ❖ Liabilities
- ❖ Net Cash Flow

Format each card: turn on **data label**, change color, and resize for clarity.

B. Clustered Column Chart – Revenue vs Expenses

1. Select **Clustered Column Chart**
2. Axis → Date, Values → Revenue, Operating_Expense
3. Use different colors (blue vs red), and add a chart title.

C. Line Chart – Net Income Trend

1. Insert a **Line Chart**
2. Axis → Date, Values → Net_Income
3. Format → Enable **Data Labels**, add title "Net Income Over Time"

D. Clustered Bar Chart – Assets vs Liabilities

1. Insert **Bar Chart**
2. Axis → Date, Values → Assets, Liabilities
3. Customize color coding (Assets = green, Liabilities = orange)

E. Area Chart – Equity Trend

1. Insert **Area Chart**
2. Axis → Date, Values → Equity
3. Add smooth color fill and data labels

F. Line Chart – Net Cash Flow Trend

1. Insert **Line Chart**
2. Axis → Date, Values → Net_Cash_Flow
3. Format: green for positive cash flow, red for negative

G. Stacked Column Chart – Inflows vs Outflows

1. Select **Stacked Column Chart**
2. Axis → Date
3. Values → Cash_Inflows, Cash_Outflows

H. Gauge Charts – Key Ratios

1. Use **Gauge Visual**
2. For each ratio:

- Value → Gross Margin %, Net Margin %, Debt to Equity Ratio
- Set min = 0, max = 100 (for % metrics)

I. Table or Matrix – Financial Summary

1. Select **Table** or **Matrix Visual**
2. Columns → Date, Revenue, Gross_Profit, Net_Income, Operating_Expense
3. Add conditional formatting:
 - ❖ Net Income < 0 → Red
 - ❖ Profit Margin > 50% → Green

Step 7: Add Filters & Slicers

1. Add **Slicers** for:

Year
Month
Account_Type
Department

Slicers make visuals dynamic and user-friendly.

Step 8: Dashboard Design and Layout

Suggested Layout:

Section	Visuals
Top Row	KPI Cards
2nd Row Left	Revenue vs Expense (Column Chart)
2nd Row Right	Net Income (Line Chart)
3rd Row Left	Assets vs Liabilities (Bar Chart)
3rd Row Right	Equity (Area Chart)
Bottom Left	Cash Inflows/Outflows (Stacked Column)
Bottom Right	Net Cash Flow (Line Chart)
Footer/Extra	Table, Ratios (Gauges), Filters/Slicers

Step 9: Publish and Share

1. Save file: Financial_Dashboard.pbix
2. Click **Home** → **Publish** → **My Workspace**
3. In **Power BI Service**, set:
 - ❖ **Scheduled Refresh** (if data updates)

- ❖ **Access permissions** (for finance teams)

Step 10: Advanced Enhancements

- ❖ **Drill-through:** Right-click on a year/month → see detailed breakdown
- ❖ **Forecasting:** Enable forecast for Revenue and Net Income (Line Chart → Analytics → Forecast)
- ❖ **What-if Parameters:** Create sliders to simulate 10% revenue increase or 5% expense cut

Final Dashboard Output Overview:

Section	KPIs/Visuals
Overview	KPI Cards: Revenue, Income, Assets, Liabilities, Cash
Income Statement	Column + Line: Revenue, COGS, Operating Expenses
Balance Sheet	Bar + Area: Assets, Liabilities, Equity
Cash Flow	Stacked Column + Line: Inflows, Outflows, Net Cash
Ratios	Gauge or KPI: Gross Margin, Net Margin, Debt/Equity
Detail	Tables or Matrix for monthly reports